

Approximate message passing for orthogonally invariant ensembles: multivariate non-linearities and spectral initialization

XINYI ZHONG[†], TIANHAO WANG[†] AND ZHOU FAN^{*}

Department of Statistics and Data Science, Yale University, New Haven, CT 65120, USA

^{*}Corresponding author. Email: zhou.fan@yale.edu

[Received on 18 October 2021; revised on 13 February 2024; accepted on 17 August 2024]

We study a class of Approximate Message Passing (AMP) algorithms for symmetric and rectangular spiked random matrix models with orthogonally invariant noise. The AMP iterates have fixed dimension $K \geq 1$, a multivariate non-linearity is applied in each AMP iteration, and the algorithm is spectrally initialized with K super-critical sample eigenvectors. We derive the forms of the Onsager debiasing coefficients and corresponding AMP state evolution, which depend on the free cumulants of the noise spectral distribution. This extends previous results for such models with $K = 1$ and an independent initialization. Applying this approach to Bayesian principal components analysis, we introduce a Bayes-OAMP algorithm that uses as its non-linearity the posterior mean conditional on all preceding AMP iterates. We describe a practical implementation of this algorithm, where all debiasing and state evolution parameters are estimated from the observed data, and we illustrate the accuracy and stability of this approach in simulations.

Keywords: AMP algorithms; free probability; low-rank matrix estimation; spiked random matrices.

2000 Math Subject Classification: 62E20; 62H25.

1. Introduction

In recent years, Approximate Message Passing (AMP) algorithms have been used in an increasingly diverse range of applications. These algorithms were originally derived as approximations to message passing and belief propagation algorithms for densely connected graphical models [21–23,30]. They have since been successfully adapted to perform both optimization and Bayesian inference in many problems arising in high-dimensional statistics and machine learning, and we refer to [28] for a recent review.

By design, AMP algorithms are closely tailored to distributional assumptions for the data matrices to which they are applied. These algorithms exhibit fast rates of convergence for typical realizations of such random data [34], and can achieve near-optimal estimation risk in many contexts of Bayesian inference [3, 4, 17–19, 31]. Furthermore, iterates of AMP admit an exact asymptotic distributional characterization, known as its ‘state evolution’, that is simpler than that of alternative first-order procedures. Thus, AMP has also served as a broadly useful theoretical tool for analyzing the asymptotic behavior of statistical methods [6,13,24,25,35,46,47] as well as probabilistic models [12, 15, 20, 27].

[†] These authors contributed equally.

The most common examples of AMP algorithms are tailored to data matrices with i.i.d. entries, and a line of work [7, 8, 10, 11, 16, 29] has rigorously established the validity of their state evolutions in this context. This state evolution may no longer correctly describe the iterates for non-i.i.d. data, where such AMP algorithms may also exhibit divergent behavior [43, 53]. More recently, several AMP algorithms have been developed for a broader class of random matrices that are orthogonally or unitarily invariant in law, but that can have arbitrary spectral distribution. These include the Orthogonal AMP [33, 48], Vector AMP [44, 45], Convolutional AMP [49–51], and Memory AMP [32] procedures for linear models and generalized linear models with orthogonally invariant designs, as well as a general class of AMP procedures for symmetric and rectangular orthogonally invariant ensembles studied in [14, 26, 40]. Broadly, these algorithms adapt to the more general spectral laws that may arise in such models, by either using divergence-free nonlinearities or applying Onsager corrections that are tailored to these spectral laws.

1.1 Contributions

Motivated by applications to statistical principal components analysis (PCA), our work extends the class of AMP algorithms for orthogonally invariant random matrix ensembles studied in [14, 26, 40], in two directions:

1. We extend the AMP procedures from vector-valued iterates $\mathbf{u}_t \in \mathbb{R}^n$ to matrix-valued iterates $\mathbf{U}_t \in \mathbb{R}^{n \times K}$, for an arbitrary fixed dimension $K \geq 1$. We consider such AMP algorithms that apply multivariate nonlinearities in every iteration, and derive the forms of the Onsager corrections and state evolutions for algorithms of this type.
2. Importantly, the nonlinearities need not be separable across the K dimensions. This generalization is particularly useful for PCA, where (empirically, in many domains of application) there is often joint structure across coordinates of multiple PCs, and a multivariate nonlinearity should be used to regularize estimates towards this structure [54].
3. In a model of data consisting of low-rank signal plus additive noise, we develop a method of spectral initialization for the general AMP algorithms of [26], using the sample eigenvectors or singular vectors of the data. This is analogous to the work of [39] that developed this extension for AMP algorithms when the noise has i.i.d. entries.
4. Such an extension eliminates the need for an informative initialization that is independent of the data, which is typically unavailable in practice. Our analysis shows that the AMP Onsager correction and state evolution must treat the spectral initialization separately, and they take different forms from the descriptions of [26].

The first generalization above is a more direct extension of the previous analyses in [26], and we describe these results in Section 2. The second generalization to a spectral initialization constitutes the larger technical contribution of our work, and we describe it for symmetric and rectangular matrices in Sections 3 and in [Appendices B](#) respectively. Our proof is different from that of [39], and instead follows a strategy introduced in [37] of approximating the spectral initialization by a sequence of τ linear AMP steps that converge to the sample eigenvectors, as $\tau \rightarrow \infty$.

Recent independent work of [38] has used this approach to derive also the forms of spectrally-initialized AMP algorithms corresponding to the ‘single-iterate posterior mean’ PCA procedure described in [26]; this constitutes an important case of our current results. Our results expand upon [38], analyzing instead the general AMP algorithms in [26] whose nonlinearities may be functions of all

preceding AMP iterates, and in the above context of multivariate iterates with dimension $K \geq 1$. At a technical level, we avoid the restrictive assumption imposed in [38] that all free cumulants of the noise spectral law be positive, and we use an alternative strategy for analyzing the convergence of the linear AMP iterations that leads to a different and explicit assumption of sufficiently large signal strength.

Finally, as an application of these results, we propose in Section 4 a Bayes-OAMP¹ algorithm for PCA with a Bayesian prior for the PCs. This algorithm differs from the single-iterate posterior mean procedure that was analyzed in [26], computing instead the Bayes posterior mean based on the multivariate Gaussian joint law of all preceding AMP iterates, as described by the above state evolution. We demonstrate in Section 4 that this can yield a sizeable improvement in estimation accuracy for rotationally invariant noise ensembles in settings of weak signal strength.

The development and rigorous characterization of Bayes-optimal estimation procedures for this PCA problem is an interesting open question. Following the initial posting of our work, [5] has obtained the first results in this direction, deriving a conjectural form of the Bayes-optimal error in a symmetric rank-1 spiked model using the replica method, for certain examples of rotationally-invariant noise matrices defined by polynomial potentials. The authors of [5] suggested also two AMP methods that numerically attain the conjectured Bayes-optimal error in these examples, one of which is an extension of Bayes-OAMP (dubbed ‘AMP-AP’) that alternates between posterior mean and identity nonlinearities. Our current results establish the rigorous state evolution of both Bayes-OAMP and AMP-AP under a spectral initialization and sufficiently large signal strength, constituting a first step towards an analytic characterization of the estimation errors attained by these procedures.

NOTATIONAL CONVENTIONS. For random variables X and Y , $X \perp\!\!\!\perp Y$ denotes that they are independent. $\|\cdot\|$ denotes the ℓ_2 -norm for vectors and the $\ell_2 \rightarrow \ell_2$ operator norm for matrices. $\|\cdot\|_F$ is the Frobenius norm for matrices. $\text{diag}(v) \in \mathbb{R}^{K \times K}$ is the diagonal matrix with $v \in \mathbb{R}^K$ on its diagonal, and we write $\text{diag}(v) \in \mathbb{R}^{K \times K'}$ to indicate this matrix right-padded by $K' - K$ columns of 0. We adopt the convention $M^0 = \text{Id}$ for the 0th power of any square matrix M . For a block matrix $M \in \mathbb{R}^{IK \times IK}$ and $\kappa \in \mathbb{R}^{K \times K}$, $M \odot \kappa \in \mathbb{R}^{IK \times IK}$ and $\kappa \oslash M \in \mathbb{R}^{IK \times IK}$ denote the block-wise right- and left- multiplication by κ .

For a matrix $\mathbf{U} \in \mathbb{R}^{n \times K}$ and random vector $U \in \mathbb{R}^K$, we write $\mathbf{U} \xrightarrow{w_2} U$ for the Wasserstein-2 convergence of the empirical distribution of rows of $\mathbf{U}$, as $n \rightarrow \infty$. Letting u_i be the i th row of $\mathbf{U}$, this means

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{i=1}^n g(u_i) = \mathbb{E}[g(U)]$$

for any continuous function $g : \mathbb{R}^K \rightarrow \mathbb{R}$ such that $|g(u)| \leq C(1 + \|u\|^2)$ for a constant $C > 0$. We write $\langle \mathbf{U} \rangle = n^{-1} \sum_{i=1}^n u_i \in \mathbb{R}^K$ for the empirical average of rows of $\mathbf{U}$.

2. Symmetric AMP with independent initialization

In this section, we first describe extensions of the AMP algorithms and state evolution characterizations of [26] from vector-valued to matrix-valued iterates, for orthogonally invariant matrices and an independent initialization. We will then discuss signal-plus-noise models and spectral initializations in

¹ This is different from the algorithm called OAMP in [33]. Throughout the paper, we refer to our algorithm specifically as Bayes-OAMP to avoid potential confusion.

Section 3. We focus on the setting of symmetric matrices in the main text for ease of presentation, and corresponding results for rectangular matrices are given in [Appendices A and B](#).

Let $\mathbf{W} \in \mathbb{R}^{n \times n}$ be a symmetric matrix, with eigen-decomposition $\mathbf{W} = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{O}$ where $\mathbf{\Lambda} = \text{diag}(\lambda)$. We will assume that $\mathbf{O}$ is a Haar-distributed orthogonal basis of eigenvectors, so $\mathbf{W}$ is orthogonally invariant in law.

For fixed dimensions $J \geq 0$ and $K \geq 1$, consider a possible additional matrix $\mathbf{E} \in \mathbb{R}^{n \times J}$ of ‘side information’, and a sequence of Lipschitz functions $u_2, u_3, \dots$ where each $u_{t+1} : \mathbb{R}^{tK+J} \rightarrow \mathbb{R}^K$. (We may set $J = 0$ if there is no such side information.) We consider an AMP algorithm with initialization $\mathbf{U}_1 \in \mathbb{R}^{n \times K}$ independent of $\mathbf{W}$, having the iterates

$$\mathbf{Z}_t = \mathbf{W}\mathbf{U}_t - \mathbf{U}_1 b_{t1}^\top - \mathbf{U}_2 b_{t2}^\top - \dots - \mathbf{U}_t b_{tt}^\top \quad (2.1)$$

$$\mathbf{U}_{t+1} = u_{t+1}(\mathbf{Z}_1, \dots, \mathbf{Z}_t, \mathbf{E}). \quad (2.2)$$

Here $b_{ts} \in \mathbb{R}^{K \times K}$ is a matrix-valued Onsager debiasing coefficient for each $s = 1, \dots, t$, and $u_{t+1}(\cdot)$ is applied row-wise to $(\mathbf{Z}_1, \dots, \mathbf{Z}_t, \mathbf{E}) \in \mathbb{R}^{n \times (tK+J)}$ to yield each next iterate $\mathbf{U}_{t+1} \in \mathbb{R}^{n \times K}$.

DEBIASING COEFFICIENTS. Let $\partial_s u_{t+1}(\mathbf{Z}_1, \dots, \mathbf{Z}_t, \mathbf{E}) \in \mathbb{R}^{K \times K}$ denote the Jacobian of u_{t+1} in its vector argument $\mathbf{Z}_s$, which exists Lebesgue-a.e. since $u_{t+1}(\cdot)$ is Lipschitz [55, Theorem 2.2.1]. Denote

$$\langle \partial_s \mathbf{U}_{t+1} \rangle = \frac{1}{n} \sum_{i=1}^n \partial_s u_{t+1}(z_{1,i}, \dots, z_{t,i}, e_i)$$

where $z_{t,i} \in \mathbb{R}^K$ and $e_i \in \mathbb{R}^J$ are the i^{th} rows of $\mathbf{Z}_t$ and $\mathbf{E}$. For each $T \geq 1$, define the $TK \times TK$ block-lower-triangular matrix

$$\boldsymbol{\phi}_T = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \langle \partial_1 \mathbf{U}_2 \rangle & 0 & \cdots & 0 & 0 \\ \langle \partial_1 \mathbf{U}_3 \rangle & \langle \partial_2 \mathbf{U}_3 \rangle & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \langle \partial_1 \mathbf{U}_T \rangle & \langle \partial_2 \mathbf{U}_T \rangle & \cdots & \langle \partial_{T-1} \mathbf{U}_T \rangle & 0 \end{pmatrix}. \quad (2.3)$$

Let Λ be a random variable on $\mathbb{R}$ with compact support, which will be the limit eigenvalue distribution of $\mathbf{W}$ as $n \rightarrow \infty$. Let $\{\kappa_j\}_{j \geq 1}$ be the free cumulants of Λ —see e.g. [26, Section 2.3] for definitions. Applying the convention $\boldsymbol{\phi}_t^0 = \text{Id}$, we take the debiasing coefficient matrices $\{b_{ts}\}$ in (2.1) up to iteration T to be the blocks of

$$\mathbf{b}_T = \sum_{j=0}^{\infty} \kappa_{j+1} \boldsymbol{\phi}_T^j \stackrel{\text{def}}{=} \begin{pmatrix} b_{11} & & & & \\ b_{21} & b_{22} & & & \\ \vdots & \vdots & \ddots & & \\ b_{T1} & b_{T2} & \cdots & b_{TT} & \end{pmatrix} \in \mathbb{R}^{TK \times TK}.$$

This may be interpreted as the R -transform of Λ applied to $\boldsymbol{\phi}_T$ (cf. Section 3.1). Note that this is a finite sum, because $\boldsymbol{\phi}_T^j = 0$ for all $j \geq T$. We have $b_{tt} = \kappa_1 \text{Id}$ for every $t \geq 1$, which vanishes if Λ has mean $\kappa_1 = 0$.

STATE EVOLUTION. This choice of debiasing coefficients leads to the empirical distribution of rows of $(\mathbf{Z}_1, \dots, \mathbf{Z}_T)$ having an asymptotically mean-zero multivariate Gaussian limit, as $n \rightarrow \infty$ for any fixed iteration T . The limit Gaussian law is described by its covariance matrix $\Sigma_T \in \mathbb{R}^{TK \times TK}$, which may be defined recursively via the following state evolution:

Let (U_1, E) be an initial random vector with $U_1 \in \mathbb{R}^K$ and $E \in \mathbb{R}^J$, representing the limit empirical distribution of rows of $(\mathbf{U}_1, \mathbf{E})$. Inductively for $t = 1, 2, 3, \dots$, having defined the joint law of $(U_1, \dots, U_t, Z_1, \dots, Z_{t-1}, E)$, define the $tK \times tK$ matrices

$$\Delta_t = \begin{pmatrix} \mathbb{E}[U_1 U_1^\top] & \mathbb{E}[U_1 U_2^\top] & \cdots & \mathbb{E}[U_1 U_t^\top] \\ \mathbb{E}[U_2 U_1^\top] & \mathbb{E}[U_2 U_2^\top] & \cdots & \mathbb{E}[U_2 U_t^\top] \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[U_t U_1^\top] & \mathbb{E}[U_t U_2^\top] & \cdots & \mathbb{E}[U_t U_t^\top] \end{pmatrix}, \quad (2.4)$$

$$\Phi_t = \begin{pmatrix} 0 & 0 & \cdots & 0 & 0 \\ \mathbb{E}[\partial_1 u_2(Z_1, E)] & 0 & \cdots & 0 & 0 \\ \mathbb{E}[\partial_1 u_3(Z_1, Z_2, E)] & \mathbb{E}[\partial_2 u_3(Z_1, Z_2, E)] & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \mathbb{E}[\partial_1 u_t(Z_1, \dots, Z_{t-1}, E)] & \mathbb{E}[\partial_2 u_t(Z_1, \dots, Z_{t-1}, E)] & \cdots & \mathbb{E}[\partial_{t-1} u_t(Z_1, \dots, Z_{t-1}, E)] & 0 \end{pmatrix}. \quad (2.5)$$

Here, Φ_t corresponds to the large- n limit of ϕ_t defined in (2.3). Then define the covariance Σ_t by

$$\Sigma_t = \sum_{j=0}^{\infty} \Theta^{(j)}[\Phi_t, \kappa_{j+2} \Delta_t] \quad \text{where} \quad \Theta^{(j)}[\Phi, \kappa \Delta] = \sum_{i=0}^j \Phi^i (\kappa \Delta) (\Phi^\top)^{j-i}. \quad (2.6)$$

Define the next joint law of $(U_1, \dots, U_{t+1}, Z_1, \dots, Z_t, E)$ by

$$(Z_1, \dots, Z_t) \sim \mathcal{N}(0, \Sigma_t) \perp (U_1, E), \quad U_{s+1} = u_{s+1}(Z_1, \dots, Z_s, E) \text{ for each } s = 1, \dots, t. \quad (2.7)$$

Under these inductive definitions, it may be checked that the upper-left $(t-1) \times (t-1)$ blocks of Σ_t coincide with Σ_{t-1} .

This state evolution characterizes the iterates of the AMP algorithm (2.1–2.2), under the following assumptions:

Assumption 2.1. The matrix $\mathbf{W} = \mathbf{O}^\top \text{diag}(\lambda) \mathbf{O}$ and random variable Λ satisfy

1. $\mathbf{O}$ is random and Haar-distributed over the orthogonal group.
2. λ is independent of $\mathbf{O}$, and its empirical distribution converges weakly a.s. to Λ as $n \rightarrow \infty$.
3. Λ has compact support $\text{supp}(\Lambda)$. Denoting $(\lambda_-, \lambda_+) = (\min \text{supp}(\Lambda), \max \text{supp}(\Lambda))$, we have $\min(\lambda) \rightarrow \lambda_-$ and $\max(\lambda) \rightarrow \lambda_+$ a.s. as $n \rightarrow \infty$.

Assumption 2.2. The AMP initialization $\mathbf{U}_1$, functions $u_2, u_3, \dots$, and random vectors (U_1, E) satisfy

1. $(\mathbf{U}_1, \mathbf{E}) \in \mathbb{R}^{n \times (K+J)}$ is independent of $\mathbf{O}$, and $(\mathbf{U}_1, \mathbf{E}) \xrightarrow{W_2} (U_1, E)$ a.s. as $n \rightarrow \infty$.
2. Each $u_{t+1}(\cdot)$ is Lipschitz in all arguments. For each $s = 1, \dots, t$, $\partial_s u_{t+1}(Z_1, \dots, Z_t, E)$ exists and is continuous on a set of probability 1 under the law of $(Z_1, \dots, Z_t, E)$ defined by (2.7).

THEOREM 2.3. Suppose Assumptions 2.1 and 2.2 hold. For any $T \geq 1$, consider the AMP algorithm (2.1–2.2) up to iteration T , and define $(U_1, \dots, U_{T+1}, Z_1, \dots, Z_T, E)$ by the state evolution (2.7). Then almost surely as $n \rightarrow \infty$,

$$(\mathbf{U}_1, \dots, \mathbf{U}_{T+1}, \mathbf{Z}_1, \dots, \mathbf{Z}_T, \mathbf{E}) \xrightarrow{W_2} (U_1, \dots, U_{T+1}, Z_1, \dots, Z_T, E).$$

The proof of Theorem 2.3 is an extension of that of [26, Theorem 4.3 and Corollary 4.4]. Compared with [26, Corollary 4.4], Theorem 2.3 considers matrix-valued iterates having dimension $n \times K$, relaxes the needed convergence $(\mathbf{U}_1, \mathbf{E}) \rightarrow (U_1, E)$ from Wasserstein- p for all orders $p \geq 1$ to only Wasserstein-2, and relaxes the continuous-differentiability requirement for each function $u_{t+1}(\cdot)$ to the weaker condition of Assumption 2.2(b). We describe the modifications of the proofs of [26] needed to establish Theorem 2.3 in Appendix F.

REMARK 2.1. We have defined b_{ts} in (2.1) using the free cumulants of the limit spectral distributions. Theorem 2.3 then also holds for any AMP algorithm where b_{ts} are replaced by b'_{ts} such that $\|b_{ts} - b'_{ts}\| \rightarrow 0$ a.s. as $n \rightarrow \infty$. In particular, they hold if b_{ts} are defined with $\{\kappa_j\}$ replaced by consistent estimates of these limit free cumulants.

3. Spectral initialization for the symmetric spiked model

We now develop versions of the preceding AMP algorithms for ‘spiked’ signal-plus-noise models, with spectral initialization. Consider a rank- K' symmetric spiked model

$$\mathbf{X} = \sum_{k=1}^{K'} \frac{\theta_k}{n} \mathbf{u}_*^k \mathbf{u}_*^{k\top} + \mathbf{W} \in \mathbb{R}^{n \times n}, \quad (3.1)$$

where $\mathbf{u}_*^1, \dots, \mathbf{u}_*^{K'}$ are K' orthogonal ‘signal’ eigenvectors, $\theta_1, \dots, \theta_{K'}$ are non-zero signal eigenvalues, and the noise matrix $\mathbf{W} = \mathbf{O}^\top \mathbf{A} \mathbf{O}$ is symmetric and rotationally-invariant in law. We distinguish this rank K' from the dimension K of the AMP iterates, in anticipation of applications where K' may be modeled as large, and the practitioner may wish to apply AMP to estimate small subsets of K signal eigenvectors at a time. We develop a version of AMP where any K super-critical sample eigenvectors of $\mathbf{X}$ may be chosen as the spectral initialization.

In the model (3.1), we fix the normalization

$$\|\mathbf{u}_*^k\|^2 = n, \quad \mathbf{u}_*^{j\top} \mathbf{u}_*^k = 0 \quad \text{for all } j \neq k \in \{1, \dots, K'\}. \quad (3.2)$$

We order the signal components such that the first K will correspond to the spectral initialization, and the remaining $K' - K$ are ordered arbitrarily. (Thus, $\theta_1, \dots, \theta_{K'}$ are not sorted, and they may have arbitrary signs.) Supposing that K_+ values $\theta_1, \dots, \theta_{K'}$ are positive and $K_- = K' - K_+$ values are negative, we

denote by

$$\lambda_1(\mathbf{X}), \dots, \lambda_{K'}(\mathbf{X}) \quad (3.3)$$

the largest K_+ and smallest K_- sample eigenvalues of $\mathbf{X}$, sorted in the same order as $\theta_1, \dots, \theta_{K'}$. We denote the associated sample eigenvectors of $\mathbf{X}$ by $\mathbf{f}_{\text{pca}}^1, \dots, \mathbf{f}_{\text{pca}}^{K'}$, with the normalization and sign convention

$$\|\mathbf{f}_{\text{pca}}^k\|^2 = n, \quad \mathbf{f}_{\text{pca}}^k \top \mathbf{u}_*^k \geq 0, \quad \mathbf{f}_{\text{pca}}^j \top \mathbf{f}_{\text{pca}}^k = 0 \quad \text{for all } j \neq k \in \{1, \dots, K'\}. \quad (3.4)$$

3.1 Preliminaries on sample eigenvectors and the R -transform

EIGENVECTORS AND SPECTRAL PHASE TRANSITION. For the signal-plus-noise model (3.1), the quantitative behavior of the leading (positive and negative) sample eigenvalues/eigenvectors of $\mathbf{X}$ and associated phenomena of spectral phase transitions were studied in [9], extending the work of [1,2,41] for models with i.i.d. noise. These depend on the Cauchy-transform of the limit spectral distribution of $\mathbf{W}$, and we briefly review these results here.

Let Λ be the limit spectral distribution of $\mathbf{W}$. Denote the Cauchy-transform of Λ by

$$G(z) = \mathbb{E}[(z - \Lambda)^{-1}] \quad \text{for } z \in (\lambda_+, \infty) \cup (-\infty, \lambda_-)$$

where $\lambda_{\pm}$ are the endpoints of support of Λ as defined in Assumption 2.1(c). $G(z)$ is strictly decreasing and positive on (λ_+, ∞) and strictly decreasing and negative on $(-\infty, \lambda_-)$, and hence admits a functional inverse $G^{-1}(z)$ on $(0, G(\lambda_+)) \cup (G(\lambda_-), 0)$ where $G(\lambda_{\pm}) = \lim_{z \rightarrow \lambda_{\pm}} G(z)$. For $k \in \{1, \dots, K'\}$ such that $1/\theta_k$ belongs to this domain of $G^{-1}(z)$, define

$$\lambda_{\text{pca},k} = G^{-1}(1/\theta_k), \quad \mu_{\text{pca},k}^2 = \frac{-1}{\theta_k^2 G'(\lambda_{k,\text{pca}})}. \quad (3.5)$$

The following theorem summarizes several results of [9, Theorems 2.1 and 2.2], which establish the first-order behavior of the leading sample eigenvalues and eigenvectors.

THEOREM 3.1 ([9]). Suppose $\mathbf{W}$ satisfies Assumption 2.1, and $\theta_1, \dots, \theta_{K'}$ are distinct and fixed as $n \rightarrow \infty$. Then for each $k \in \{1, \dots, K'\}$ where $\theta_k > 1/G(\lambda_+) \geq 0$ or $\theta_k < 1/G(\lambda_-) \leq 0$, almost surely

$$\lim_{n \rightarrow \infty} \lambda_k(\mathbf{X}) = \lambda_{\text{pca},k}, \quad \lim_{n \rightarrow \infty} \left(\frac{\mathbf{f}_{\text{pca}}^k \top \mathbf{u}_*^k}{n} \right)^2 = \mu_{\text{pca},k}^2, \quad \lim_{n \rightarrow \infty} \left(\frac{\mathbf{f}_{\text{pca}}^k \top \mathbf{u}_*^j}{n} \right)^2 = 0 \quad \text{for all } j \in \{1, \dots, K'\} \setminus \{k\}.$$

For each other $k \in \{1, \dots, K'\}$, $\lim_{n \rightarrow \infty} \lambda_k(\mathbf{X}) \in \{\lambda_-, \lambda_+\}$.

Thus, for a ‘super-critical’ signal eigenvalue θ_k exceeding the positive and negative phase transition thresholds $1/G(\lambda_+)$ and $1/G(\lambda_-)$, the corresponding sample eigenvalue $\lambda_k(\mathbf{X})$ converges to a deterministic value $\lambda_{\text{pca},k}$ outside the interval $[\lambda_-, \lambda_+]$, the sample eigenvector $\mathbf{f}_{\text{pca}}^k$ achieves asymptotically non-vanishing alignment with its corresponding signal vector $\mathbf{u}_*^k$, and it has asymptotically 0 alignment with the other signal vectors $\mathbf{u}_*^j$. For a ‘sub-critical’ signal eigenvalue θ_k below these phase transition

thresholds, $\lambda_k(\mathbf{X})$ converges to the spectral edges $\lambda_{\pm}$ of the noise spectral distribution. Note that $G(\lambda_+)$ and $G(\lambda_-)$ may be infinite, in which case the phase transition thresholds are 0, and all signal eigenvalues are super-critical. Whether this occurs depends on the rates of decay of the distribution of Λ at its spectral edges $\lambda_{\pm}$, and this is discussed further in [9, Proposition 2.4].

R-TRANSFORM. The above first-order limits may be re-expressed via the R -transform and free cumulants of Λ , which linearize free addition of independent random matrices. Define

$$R(z) = G^{-1}(z) - \frac{1}{z}$$

on the same domain as G^{-1} . Differentiating on both sides yields

$$R'(z) = \frac{1}{G'(G^{-1}(z))} + \frac{1}{z^2}.$$

For z small enough, $R(z)$ and its derivative have the convergent series expansions defined through the free cumulants $\{\kappa_j\}_{j \geq 1}$ of Λ ,

$$R(z) = \sum_{j=0}^{\infty} \kappa_{j+1} z^j, \quad R'(z) = \sum_{j=0}^{\infty} (j+1) \kappa_{j+2} z^j, \quad (3.6)$$

see e.g. [36, Theorem 17]. Thus, for sufficiently large $|\theta_k|$, the limits $\lambda_{\text{pca},k}$ and $\mu_{\text{pca},k}^2$ in (3.5) also have the convergent series forms

$$\lambda_{\text{pca},k} = \theta_k + R(1/\theta_k) = \theta_k + \sum_{j=0}^{\infty} \frac{\kappa_{j+1}}{\theta_k^j}, \quad 1 - \mu_{\text{pca},k}^2 = \frac{1}{\theta_k^2} R'(1/\theta_k) = \sum_{j=0}^{\infty} (j+1) \frac{\kappa_{j+2}}{\theta_k^{j+2}}. \quad (3.7)$$

3.2 AMP algorithm

Isolating the first $K \leq K'$ of the (unsorted) signal components for the spectral initialization, denote

$$S = \text{diag}(\theta_1, \dots, \theta_K) \in \mathbb{R}^{K \times K}, \quad S' = \text{diag}(\theta_1, \dots, \theta_{K'}) \in \mathbb{R}^{K' \times K'},$$

$$\mathbf{F}_{\text{pca}} = (\mathbf{f}_{\text{pca}}^1, \dots, \mathbf{f}_{\text{pca}}^K) \in \mathbb{R}^{n \times K}, \quad \mathbf{U}'_* = (\mathbf{u}_*^1, \dots, \mathbf{u}_*^{K'}) \in \mathbb{R}^{n \times K'}.$$

Here we assume that $\theta_1, \dots, \theta_K$ are known for simplicity. The following procedure is also applicable when $\theta_1, \dots, \theta_K$ are unknown, as one can consistently estimate these values using Theorem 3.1, and we discuss this estimation in Section 4.2. We consider an AMP algorithm with iterates of dimension $n \times K$, initialized spectrally at

$$\mathbf{U}_0 = \mathbf{F}_{\text{pca}} S^{-1}, \quad \mathbf{F}_0 = \mathbf{F}_{\text{pca}}. \quad (3.8)$$

For a sequence of Lipschitz functions $u_1, u_2, \dots$ where $u_t : \mathbb{R}^{tK} \rightarrow \mathbb{R}^K$, this algorithm then iteratively computes for $t \geq 1$

$$\mathbf{U}_t = u_t(\mathbf{F}_0, \mathbf{F}_1, \dots, \mathbf{F}_{t-1}), \quad (3.9)$$

$$\mathbf{F}_t = \mathbf{X}\mathbf{U}_t - \mathbf{U}_0 b_{t0}^\top - \mathbf{U}_1 b_{t1}^\top - \dots - \mathbf{U}_t b_{tt}^\top. \quad (3.10)$$

Thus, each $u_t(\cdot)$ may depend on the preceding iterates $\mathbf{F}_0, \dots, \mathbf{F}_{t-1}$, including the spectral initialization. An additional matrix of side information may be incorporated into each $u_t(\cdot)$ as in (2.2), but we omit this here for simplicity. To ease notation in the analysis, we have shifted the initialization index from 1 to 0.

DEBIASING COEFFICIENTS. For each fixed $s \geq 1$, define diagonal matrices $\tilde{\kappa}_s, \hat{\kappa}_s \in \mathbb{R}^{K \times K}$ by the matrix series

$$\tilde{\kappa}_s = \sum_{j=0}^{\infty} \kappa_{j+s} S^{-j}, \quad \hat{\kappa}_s = \sum_{j=0}^{\infty} (j+1) \kappa_{j+s} S^{-j}. \quad (3.11)$$

For each $T \geq 1$, define the block-lower-triangular matrix

$$\boldsymbol{\phi}_T = \left(\langle \partial_s \mathbf{U}_r \rangle \right)_{r,s \in \{0, \dots, T\}} \in \mathbb{R}^{(T+1)K \times (T+1)K}$$

with row blocks indexed by r and column blocks by s , where $\langle \partial_s \mathbf{U}_r \rangle = n^{-1} \sum_{i=1}^n \partial_s u_r(f_{0,i}, \dots, f_{r-1,i})$, $\partial_s u_r \in \mathbb{R}^{K \times K}$ denotes the Jacobian of $u_r(f_0, \dots, f_{r-1})$ in the argument f_s , and $\partial_s u_r = 0$ for $s \geq r$. Define the matrix series

$$\mathbf{b}_T = \sum_{j=0}^{\infty} \kappa_{j+1} \boldsymbol{\phi}_T^j, \quad \tilde{\mathbf{b}}_T = \sum_{j=0}^{\infty} \boldsymbol{\phi}_T^j \odot \tilde{\kappa}_{j+1},$$

where we recall our notation $M \odot \tilde{\kappa}_s$ and $\tilde{\kappa}_s \odot M$ for the right- and left- multiplication of each block of M . Indexing blocks by $\{0, \dots, T\}$ and writing $[t, s]$ to denote the $K \times K$ submatrix corresponding to row block t and column block s , we set the debiasing coefficients of (3.10) up to iteration T as

$$b_{ts} = \begin{cases} \tilde{\mathbf{b}}_T[t, s] & \text{if } s = 0, \\ \mathbf{b}_T[t, s] & \text{otherwise.} \end{cases} \quad (3.12)$$

STATE EVOLUTION. The state of this algorithm up to iteration T is characterized by

$$\boldsymbol{\mu}_T = \begin{pmatrix} \mu_0 \\ \vdots \\ \mu_T \end{pmatrix} \in \mathbb{R}^{(T+1)K \times K'}, \quad \boldsymbol{\Sigma}_T = \begin{pmatrix} \sigma_{00} & \cdots & \sigma_{0T} \\ \vdots & \ddots & \vdots \\ \sigma_{T0} & \cdots & \sigma_{TT} \end{pmatrix} \in \mathbb{R}^{(T+1)K \times (T+1)K}$$

and a corresponding joint law for random vectors $(U'_*, U_0, \dots, U_{T+1}, F_0, \dots, F_T)$, where U'_*, U_t, F_t represent the limit distributions of $\mathbf{U}'_*, \mathbf{U}_t, \mathbf{F}_t$, the matrix $\boldsymbol{\Sigma}_T$ is the covariance of $(F_0, \dots, F_T)$ conditional

on U'_* and μ_T relates the conditional mean of $(F_0, \dots, F_T)$ to U'_* . These are defined recursively as follows:

Let $U'_* \in \mathbb{R}^{K'}$ be a random vector satisfying $\mathbb{E}[U'_* U'^\top_*] = \text{Id}$, representing the limit distribution of rows of $\mathbf{U}'_*$ under the normalization (3.2). Set $\mu_{\text{pca}} = \text{diag}(\mu_{\text{pca},1}, \dots, \mu_{\text{pca},K}) \in \mathbb{R}^{K \times K'}$, where the last $K' - K$ columns are 0. Then $\text{Id} - \mu_{\text{pca}} \mu_{\text{pca}}^\top = \text{diag}(1 - \mu_{\text{pca},1}^2, \dots, 1 - \mu_{\text{pca},K}^2) \in \mathbb{R}^{K \times K}$. We initialize

$$\mu_0 = \mu_0 = \mu_{\text{pca}}, \quad \Sigma_0 = \sigma_{00} = \text{Id} - \mu_{\text{pca}} \mu_{\text{pca}}^\top.$$

Inductively, having defined $(\mu_{t-1}, \Sigma_{t-1})$, we define a joint law for $(U'_*, U_0, \dots, U_t, F_0, \dots, F_{t-1})$ by

$$\begin{aligned} (F_0, \dots, F_{t-1}) \mid U'_* &\sim \mathcal{N}(\mu_{t-1} \cdot U'_*, \Sigma_{t-1}), \\ U_0 &= S^{-1} F_0, \quad U_s = u_s(F_0, \dots, F_{s-1}) \text{ for } s = 1, \dots, t. \end{aligned} \quad (3.13)$$

We then define the next mean transformation μ_t to have the blocks

$$\mu_s = \mathbb{E}[U_s U'^\top_*] \cdot S' \text{ for each } s = 0, \dots, t. \quad (3.14)$$

For $s = 0$, this may be checked to coincide with the above initialization μ_{pca} .

We decompose the second moment matrix of $(U_0, \dots, U_t)$ into four parts:

$$\begin{aligned} \Delta_t \stackrel{\text{def}}{=} & \underbrace{\begin{pmatrix} 0 & 0 & \dots & 0 \\ 0 & \mathbb{E}[U_1 U_1^\top] & \dots & \mathbb{E}[U_1 U_t^\top] \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \mathbb{E}[U_t U_1^\top] & \dots & \mathbb{E}[U_t U_t^\top] \end{pmatrix}}_{\bar{\Delta}_t} + \underbrace{\begin{pmatrix} 0 & \mathbb{E}[U_0 U_1^\top] & \dots & \mathbb{E}[U_0 U_t^\top] \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}}_{\tilde{\Delta}_t} \\ & + \underbrace{\begin{pmatrix} 0 & 0 & \dots & 0 \\ \mathbb{E}[U_1 U_0^\top] & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ \mathbb{E}[U_t U_0^\top] & 0 & \dots & 0 \end{pmatrix}}_{\tilde{\Delta}_t^\top} + \underbrace{\begin{pmatrix} \mathbb{E}[U_0 U_0^\top] & 0 & \dots & 0 \\ 0 & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 \end{pmatrix}}_{\hat{\Delta}_t}, \end{aligned} \quad (3.15)$$

set

$$\Delta_t^{(j)} = \kappa_{j+2} \bar{\Delta}_t + \tilde{\kappa}_{j+2} \odot \tilde{\Delta}_t + \tilde{\Delta}_t^\top \odot \tilde{\kappa}_{j+2} + \hat{\kappa}_{j+2} \odot \hat{\Delta}_t,$$

and define analogously to (2.3)

$$\Phi_t = \left(\mathbb{E}[\partial_s u_r(F_0, \dots, F_{r-1})] \right)_{r,s \in \{0, \dots, t\}} \in \mathbb{R}^{(t+1)K \times (t+1)K}.$$

Then, recalling the function $\Theta^{(j)}[\cdot, \cdot]$ from (2.6), we define the next covariance matrix Σ_t by

$$\Sigma_t = \sum_{j=0}^{\infty} \Theta^{(j)}[\Phi_t, \Delta_t^{(j)}]. \quad (3.16)$$

It may be checked from these definitions that the first t blocks of μ_t coincide with μ_{t-1} , and the upper-left $t \times t$ blocks of Σ_t coincide with Σ_{t-1} .

Our main result in the context of model (3.1) shows that this state evolution provides a rigorous characterization of the AMP algorithm (3.9–3.10) with the spectral initialization (3.8), under the following assumptions:

Assumption 3.2.

- (a) $\mathbf{U}_* = (\mathbf{u}_*^1, \dots, \mathbf{u}_*^{K'})$ is independent of $\mathbf{O}$, satisfies (3.2), and $\mathbf{U}_* \xrightarrow{W_2} U'_*$ a.s. as $n \rightarrow \infty$ where $\mathbb{E}[U'_* U'^*] = \text{Id}$.
- (b) Each $u_{t+1}(\cdot)$ is Lipschitz in all arguments. For each $s = 0, \dots, t$ and all (μ, Σ) in a sufficiently small open neighborhood of (μ_t, Σ_t) defined by (3.14) and (3.16), $\partial_s u_{t+1}(F_0, \dots, F_t)$ exists and is continuous on a set of probability 1 under the marginal law of $(F_0, \dots, F_t)$ defined by $(F_0, \dots, F_t) \mid U'_* \sim \mathcal{N}(\mu \cdot U'_*, \Sigma)$.
- (c) $\theta_1, \dots, \theta_{K'}$ are distinct. For each $k \in \{1, \dots, K\}$, either $\theta_k > G(1/\lambda_+) \geq 0$ or $\theta_k < G(1/\lambda_-) \leq 0$, and there exists some constant $\iota \in (0, 1)$ such that

$$\frac{\max(|\lambda_+|, |\lambda_-|)}{|\theta_k|} + \sum_{j=1}^{\infty} \frac{|\kappa_j|}{|\theta_k|^j \cdot j^{j-1}} < 1. \quad (3.17)$$

Assumption 3.2(c) requires $\theta_1, \dots, \theta_K$ for the first K selected signals to be ‘super-critical’, as described by Theorem 3.1. Furthermore, (3.17) requires each $|\theta_k|$ to exceed some constant depending only on the law of Λ . (We believe that this additional requirement may be an artifact of the proof technique, and we do not optimize the value of this constant.) Note that this requirement (3.17) is sufficient to imply that the series (3.6) for $R(z)$ and $R'(z)$ are absolutely convergent at $z = 1/\theta_k$, and hence the series (3.11) defining $\tilde{\kappa}_s, \hat{\kappa}_s$ are also absolutely convergent.

We remark that we assume for simplicity the distinctness of $\theta_1, \dots, \theta_{K'}$. If these signal values are not all distinct, then the AMP state evolution parameters may not concentrate around deterministic values, and this has been discussed and analyzed in [39] when the noise matrix is Gaussian. The study of similar phenomena in our model with rotationally-invariant noise may also be of interest, but we will not pursue this direction in the current paper.

THEOREM 3.3. Consider the symmetric spiked model (3.1), where Assumptions 2.1 and 3.2 hold. For any $T \geq 1$, consider the spectrally initialized AMP algorithm (3.8–3.10) up to iteration T , and define $(U'_*, U_0, \dots, U_{T+1}, F_0, \dots, F_T)$ by (3.13). Then almost surely as $n \rightarrow \infty$,

$$(\mathbf{U}'_*, \mathbf{U}_0, \dots, \mathbf{U}_{T+1}, \mathbf{F}_0, \dots, \mathbf{F}_T) \xrightarrow{W_2} (U'_*, U_0, \dots, U_{T+1}, F_0, \dots, F_T).$$

REMARK 3.1. As in Remark 2.1, we have defined b_{ts} in (3.12) using the free cumulants $\{\kappa_j\}$ of the limit noise spectral distribution, as well as the true signal values $\theta_1, \dots, \theta_K$. Theorem 3.3 then also holds when

b_{ts} are replaced by b'_{ts} such that $\|b_{ts} - b'_{ts}\| \rightarrow 0$ a.s., and in particular if $\{\kappa_j\}$, $\{\tilde{\kappa}_j\}$, $\{\hat{\kappa}_j\}$, and $\theta_1, \dots, \theta_K$ are replaced by consistent estimates of these quantities.

3.3 Proof idea

We provide the proof of Theorem 3.3 in Appendix D, and describe here the main idea. We construct an auxiliary AMP algorithm consisting of two phases. For some $\tau \geq 1$, we index the iterates in the first phase from $-\tau$ to 0. We consider an initialization $\mathbf{U}_{-\tau}^{(\tau)}$ that is independent of $\mathbf{W}$, and apply linear AMP iterations with $\mathbf{U}_t^{(\tau)} = \mathbf{F}_{t-1}^{(\tau)} S^{-1}$ for $t = -\tau + 1, \dots, 0$. This has the effect of implementing a version of the power method to compute the sample eigenvectors $\mathbf{F}_{\text{pca}}$ of $\mathbf{X}$. A main step of the proof is to show that as $\tau \rightarrow \infty$, the final iterate $\mathbf{F}_0^{(\tau)}$ obtained from these linear updates approaches the spectral initialization $\mathbf{F}_0 = \mathbf{F}_{\text{pca}}$. Here, the assumption in (3.17) guarantees the desired convergence of $\mathbf{F}_0^{(\tau)}$ and the corresponding state evolution. Our analysis directly characterizes the alignment between $\mathbf{F}_0^{(\tau)}$ and $\mathbf{F}_{\text{pca}}$ (which then implies the convergence of the state evolution) under general spectral distributions. This is different from [38], which studied directly the asymptotic state evolution, and instead established the optimal signal-to-noise condition for convergence of the linear AMP iterations under a more restrictive class of spectral distributions having positive free cumulants.

In the second phase, starting from $\mathbf{F}_0^{(\tau)}$, we then apply the same functions $u_t(\cdot)$ as in the spectrally-initialized AMP algorithm for $t = 1, 2, \dots$. The combined auxiliary AMP algorithm can thus be summarized as follows:

$$\mathbf{F}_t^{(\tau)} = \mathbf{X} \mathbf{U}_t^{(\tau)} - \sum_{s=-\tau}^t \mathbf{U}_s^{(\tau)} b_{ts}^{(\tau)\top}, \quad \mathbf{U}_{t+1}^{(\tau)} = u_{t+1}(\mathbf{F}_{-\tau}^{(\tau)}, \dots, \mathbf{F}_t^{(\tau)})$$

where

$$u_{t+1}(\mathbf{F}_{-\tau}^{(\tau)}, \dots, \mathbf{F}_t^{(\tau)}) = \begin{cases} \mathbf{F}_t^{(\tau)} S^{-1} & \text{for } -\tau + 1 \leq t < 0, \\ u_{t+1}(\mathbf{F}_0^{(\tau)}, \dots, \mathbf{F}_t^{(\tau)}) & \text{for } 0 \leq t \leq T. \end{cases} \quad (3.18)$$

This AMP algorithm may be analyzed using Theorem 2.3 with side information $\mathbf{E} = \mathbf{U}_*$, to yield a state evolution for iterates $t \geq 0$ characterized by some $(\boldsymbol{\mu}_t^{(\tau)}, \boldsymbol{\Sigma}_t^{(\tau)})$, which we describe explicitly in Corollary D.1. Then we prove that in the limit $\tau \rightarrow \infty$, the iterates of this auxiliary AMP algorithm for $t \geq 0$ will be close to those of the spectrally-initialized AMP algorithm, in the sense that

$$\lim_{\tau \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{\|\mathbf{F}_t^{(\tau)} - \mathbf{F}_t\|_F}{\sqrt{n}} = 0, \quad \lim_{\tau \rightarrow \infty} \limsup_{n \rightarrow \infty} \frac{\|\mathbf{U}_t^{(\tau)} - \mathbf{U}_t\|_F}{\sqrt{n}} = 0.$$

We also prove that as $\tau \rightarrow \infty$, the state evolution characterizing (3.18) converges to the state evolution described by (3.14) and (3.16), i.e. for each fixed $t \geq 1$,

$$\lim_{\tau \rightarrow \infty} \boldsymbol{\mu}_t^{(\tau)} = \boldsymbol{\mu}_t \quad \text{and} \quad \lim_{\tau \rightarrow \infty} \boldsymbol{\Sigma}_t^{(\tau)} = \boldsymbol{\Sigma}_t.$$

Combining the above two facts, we then can characterize the behavior of the AMP algorithm with spectral initialization.

4. Orthogonal AMP for Bayesian PCA

Finally, we discuss in this section an application to estimating the signal vectors $\mathbf{u}_*^k$ in the preceding symmetric signal-plus-noise model in Equation (3.1). Applications for the rectangular model are deferred to Appendix C.

The distributions of $U'_* \in \mathbb{R}^{K'}$ for the row-wise limits of $\mathbf{U}'_*$ may be interpreted as Bayesian ‘priors’ for these rows. Assuming these prior distributions are known, we describe a Bayes-OAMP method in Section 4.1 that uses Bayes posterior-mean denoisers as the nonlinearities in the preceding AMP algorithms. We suggest ways of estimating the Onsager debiasing coefficients and state evolution parameters in Section 4.2, and illustrate the method in simulation in Section 4.3. In practice, the distributions of U'_* are typically also unknown, but may be estimated from the data $\mathbf{X}$ using empirical Bayes ideas. We will not consider this additional complexity here, but refer readers to [54] for an example of this approach.

Let us describe the Bayes-OAMP method in a setting where the AMP dimension K and signal rank K' satisfy $K \leq K'$, reflecting applications where K' may be large, and one may wish to estimate smaller subsets of dependent PCs at a time. In this setting, we consider the following additional assumption for the laws of U'_* .

Assumption 4.1. The last $K' - K$ coordinates of U'_* have mean 0, and are independent of the first K coordinates.

In applications, the data $\mathbf{X}$ is often centered to have row and column means approximately 0 before applying PCA, leading to PCs also having mean approximately 0. The components $\mathbf{u}_*^k$ should ideally be grouped into small subsets of dependent signals, with the signals within each subset estimated together to maximally leverage their joint structure. The above assumption of exact independence of one such subset $(1, \dots, K)$ from the remaining signals $(K + 1, \dots, K')$ is a modeling approximation for this grouping.

Assumption 4.1 ensures that when AMP is initialized with the first K signal components, its state evolution will not depend on the remaining $K' - K$ components: In the symmetric setting of Section 3, let $U_* \in \mathbb{R}^K$ be the first K coordinates of U'_* . Then it is inductively verified using Assumption 4.1 that each mean transformation $\boldsymbol{\mu}_t \in \mathbb{R}^{(t+1)K \times K'}$ has last $K' - K$ columns equal to 0, and the law of each vector F_t and U_t depends on U'_* only via U_* . We may then write the conditional law of $(F_0, \dots, F_t)$ more simply as

$$(F_0, \dots, F_t) \mid U_* \sim \mathcal{N}(\boldsymbol{\mu}_t \cdot U_*, \boldsymbol{\Sigma}_t)$$

where, with slight abuse of notation, we write $\boldsymbol{\mu}_t \in \mathbb{R}^{(t+1)K \times K}$ for its first K columns.

4.1 Bayes-OAMP

In the symmetric spiked model (3.1), under Assumption 4.1, we consider an AMP algorithm that estimates the first K components $\mathbf{U}_* \in \mathbb{R}^{n \times K}$ of $\mathbf{U}'_*$, using as its nonlinearities the posterior-mean denoising functions

$$u_{t+1}(f_0, \dots, f_t) = \mathbb{E}[U_* \mid (F_0, \dots, F_t) = (f_0, \dots, f_t)]$$

computed under the above conditional law $(F_0, \dots, F_t) \mid U_* \sim \mathcal{N}(\boldsymbol{\mu}_t \cdot U_*, \boldsymbol{\Sigma}_t)$. Explicitly, writing as shorthand $\mathbf{f}_t = (f_0, \dots, f_t) \in \mathbb{R}^{(t+1)K}$,

$$u_{t+1}(\mathbf{f}_t) = \frac{\mathbb{E}[U_* \exp(-\frac{1}{2}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*)^\top \boldsymbol{\Sigma}_t^{-1}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*))]}{\mathbb{E}[\exp(-\frac{1}{2}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*)^\top \boldsymbol{\Sigma}_t^{-1}(\mathbf{f}_t - \boldsymbol{\mu}_t \cdot U_*))]} \in \mathbb{R}^K. \quad (4.1)$$

We will refer to the AMP algorithm (3.8–3.10) using this choice of nonlinearity as *Bayes-OAMP*.

If this posterior mean function $u_{t+1}(\cdot)$ is Lipschitz (which holds e.g. if U_* has bounded support or log-concave density) and the signal strengths $|\theta_1|, \dots, |\theta_K|$ are sufficiently large, then Theorem 3.3 implies that the mean-squared-error risk of the estimate $\mathbf{U}_t$ has the asymptotic limit

$$\lim_{n \rightarrow \infty} \frac{1}{n} \|\mathbf{U}_t - \mathbf{U}_*\|_F^2 \stackrel{\text{def}}{=} \text{MSE}(\mathbf{U}_t) = \mathbb{E}[\|\mathbf{U}_* - \mathbb{E}[U_* \mid F_0, \dots, F_t]\|^2] = \text{Tr Cov}[U_* \mid F_0, \dots, F_t].$$

Thus, these errors satisfy

1. $\text{MSE}(\mathbf{U}_{t+1}) \leq \text{MSE}(\mathbf{U}_t)$, by the property $\text{Cov}[U_* \mid F_0, \dots, F_{t+1}] \preceq \text{Cov}[U_* \mid F_0, \dots, F_t]$. This implies also by the martingale property of $U_1, U_2, \dots$ that the algorithm is convergent in the sense

$$\lim_{n \rightarrow \infty} \frac{1}{n} \|\mathbf{U}_{t+1} - \mathbf{U}_t\|_F^2 = \mathbb{E}[\|U_{t+1} - U_t\|^2] = \text{MSE}(\mathbf{U}_t) - \text{MSE}(\mathbf{U}_{t+1}) \xrightarrow{t \rightarrow \infty} 0.$$

2. $\text{MSE}(\mathbf{U}_t) \leq \text{MSE}(\mathbf{F}_{\text{pca}})$ for any iterate $t \geq 1$, because

$$\text{MSE}(\mathbf{F}_{\text{pca}}) = \text{MSE}(\mathbf{F}_0) = \mathbb{E}[\|\mathbf{U}_* - F_0\|_2^2] \geq \mathbb{E}[\|\mathbf{U}_* - \mathbb{E}[U_* \mid F_0, \dots, F_t]\|_2^2] = \text{MSE}(\mathbf{U}_t).$$

REMARK 4.1. This algorithm differs from the Bayes-AMP algorithms of [39, 42] for Gaussian noise and the single-iterate Bayes-OAMP algorithm that was studied in [26, Section 3], which use instead $u_{t+1}(f_0, \dots, f_t) = \mathbb{E}[U_* \mid F_t = f_t]$ based only on the single preceding iterate F_t .

When $\mathbf{W}$ is symmetric Gaussian noise and Λ is distributed as Wigner's semicircle law, these two approaches coincide: Indeed, in this case $\kappa_2 = 1$ and $\kappa_j = 0$ for all $j \geq 3$, so (3.16) reduces to $\boldsymbol{\Sigma}_t = \boldsymbol{\Delta}_t = \mathbb{E}[UU^\top]$ where $U = (U_0, \dots, U_t)$, coinciding with the state evolution shown in [39]. For any $t \geq 1$ and $0 \leq s \leq t$, from the martingale identity

$$\mathbb{E}[U_s U_t^\top] = \mathbb{E}[U_s \mathbb{E}[U_*^\top \mid F_0, \dots, F_t]] = \mathbb{E}[\mathbb{E}[U_s U_*^\top \mid F_0, \dots, F_t]] = \mathbb{E}[U_s U_*^\top]$$

and the definition of $\boldsymbol{\mu}_t$ in (3.14), we observe that $\boldsymbol{\Sigma}_t(\mathbf{e}_t \odot S) = \boldsymbol{\mu}_t$ and $\sigma_{tt}S = \boldsymbol{\mu}_t$, where $S = \text{diag}(\theta_1, \dots, \theta_K) \in \mathbb{R}^{K \times K}$, $\mathbf{e}_t \odot S \in \mathbb{R}^{(t+1)K \times K}$ has first t row blocks equal to 0 and last row block equal to S , and $\sigma_{tt}, \boldsymbol{\mu}_t \in \mathbb{R}^{K \times K}$ denote the last blocks of $\boldsymbol{\Sigma}_t$ and $\boldsymbol{\mu}_t$. Then $\boldsymbol{\Sigma}_t^{-1} \boldsymbol{\mu}_t = \mathbf{e}_t \odot S$ and $\sigma_{tt}^{-1} \boldsymbol{\mu}_t = S$. So

(4.1) is equivalent to the single-iterate posterior mean,

$$\begin{aligned} u_{t+1}(\mathbf{f}_t) &= \frac{\mathbb{E}[U_* \exp(f_t^\top S U_* - \frac{1}{2} U_*^\top \mu_t^\top S U_*)]}{\mathbb{E}[\exp(f_t^\top S U_* - \frac{1}{2} U_*^\top \mu_t^\top S U_*)]} \\ &= \frac{\mathbb{E}[U_* \exp(-\frac{1}{2}(f_t - \mu_t U_*)^\top \sigma_{tt}^{-1}(f_t - \mu_t U_*))]}{\mathbb{E}[\exp(-\frac{1}{2}(f_t - \mu_t U_*)^\top \sigma_{tt}^{-1}(f_t - \mu_t U_*))]} = \mathbb{E}[U_* | F_t = f_t]. \end{aligned}$$

Outside of this setting where Λ has Wigner semicircle law, these two approaches are different. The state evolution of Bayes-OAMP is more involved, and we will not pursue a theoretical characterization of its fixed point in this work, as was done for single-iterate Bayes-OAMP in [26]. We observe in simulation that the above monotonicity property of $\text{MSE}(\mathbf{U}_t)$ need not hold for single-iterate Bayes-OAMP in settings of small signal strength, as was observed also in [38], and that Bayes-OAMP can substantially improve over the single-iterate approach in such settings.

4.2 Estimating the debiasing corrections and state evolution

Numerical implementations of the Bayes-OAMP algorithms require estimating the debiasing coefficients and state evolution parameters that describe the conditional laws of $(F_0, \dots, F_t)$. We describe here one approach for this estimation for the symmetric model.

We estimate the law of Λ by the observed empirical eigenvalue distribution of $\mathbf{X}$, with largest K_+ positive eigenvalues and smallest K_- negative eigenvalues removed. This estimate is weakly consistent as $n \rightarrow \infty$ by Weyl's eigenvalue interlacing inequality. We compute the empirical moments of this law, and estimate the free cumulants $\{\kappa_s\}$ through the non-crossing moment-cumulant relations, see e.g. [26, Section 2.3]. (Alternative methods for estimating the free cumulants of Λ based on power iteration have also been discussed in [32, Proposition 1] and [52, Section 3].) For each signal value θ_k , based on (3.5) and (3.7), we then estimate

$$\theta_k \text{ by } \frac{1}{G(\lambda_{\text{pca},k})}, \quad \mu_{\text{pca},k}^2 \text{ by } \frac{-1}{\theta_k^2 G'(\lambda_{\text{pca},k})}, \quad R(1/\theta_k) \text{ by } \lambda_{\text{pca},k} - \theta_k, \quad R'(1/\theta_k) \text{ by } \theta_k^2(1 - \mu_{\text{pca},k}^2)$$

where $\lambda_{\text{pca},k}$ is the observed eigenvalue of $\mathbf{X}$, and the integrals defining $G(\cdot)$ and $G'(\cdot)$ are computed using the above empirical estimate for the law of Λ . In particular, this provides the estimate of $S = \text{diag}(\theta_1, \dots, \theta_K)$. Comparing (3.6) and (3.11), we then estimate

$$\tilde{\kappa}_1 \text{ by } \text{diag}(R(1/\theta_1), \dots, R(1/\theta_K)), \quad \hat{\kappa}_2 \text{ by } \text{diag}(R'(1/\theta_1), \dots, R'(1/\theta_K)),$$

and estimate the remaining matrices $\tilde{\kappa}_s$ and $\hat{\kappa}_s$ using the following recursions derived from (3.11),

$$\begin{aligned} \tilde{\kappa}_s &= \sum_{j=0}^{\infty} \kappa_{s+j} S^{-j} = \kappa_s I + \sum_{j=0}^{\infty} \kappa_{s+1+j} S^{-j} \cdot S^{-1} = \kappa_s I + \tilde{\kappa}_{s+1} S^{-1} \Rightarrow \tilde{\kappa}_{s+1} = (\tilde{\kappa}_s - \kappa_s I) S, \\ \hat{\kappa}_s &= \sum_{j=0}^{\infty} (j+1) \kappa_{s+j} S^{-j} = \tilde{\kappa}_s + \sum_{j=0}^{\infty} (j+1) \cdot \kappa_{s+1+j} S^{-j} \cdot S^{-1} = \tilde{\kappa}_s + \hat{\kappa}_{s+1} S^{-1} \Rightarrow \hat{\kappa}_{s+1} = (\hat{\kappa}_s - \tilde{\kappa}_s) S. \end{aligned}$$

Next, explicitly differentiating (4.1) to compute the matrices $\langle \partial_s \mathbf{U}_t \rangle$ constituting $\boldsymbol{\phi}_t$, and combining with the above, we obtain consistent estimates of the debiasing coefficients b_{ts} . For the state evolution, we note that numerically evaluating the expectations defining $(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)$ may be prohibitive. We instead approximate the second-moment matrices $\mathbb{E}[U_s U_t^\top]$ by the empirical averages $n^{-1} \mathbf{U}_s^\top \mathbf{U}_t$. We approximate $\boldsymbol{\phi}_t$ also by the empirical average $\hat{\boldsymbol{\phi}}_t$, and combine with the above estimates of $\kappa_s, \tilde{\kappa}_s, \hat{\kappa}_s$ according to the definition (3.16) to obtain a consistent estimate of $\boldsymbol{\Sigma}_t$. We use the martingale identity $\mathbb{E}[U_t U_*^\top] = \mathbb{E}[U_t U_t^\top]$ for $t \geq 1$ to approximate $\mathbb{E}[U_t U_*^\top]$ by the empirical average $n^{-1} \mathbf{U}_t^\top \mathbf{U}_t$, and use this to consistently estimate $\boldsymbol{\mu}_t$.

We remark that when the AMP algorithm approaches convergence, the covariance matrix $\boldsymbol{\Sigma}_t$ becomes nearly singular, leading to potential instabilities in evaluating the posterior mean function (4.1) and its Jacobian. We use a simple early stopping rule of terminating the iterations when the smallest eigenvalue of our estimate for $\boldsymbol{\Sigma}_t$ falls below a small threshold. Alternatively, a small ridge regularization may be used when computing $\boldsymbol{\Sigma}_t^{-1}$.

4.3 Simulation studies

We compare the estimation accuracy of three AMP algorithms under various noise spectral distributions for $\mathbf{W}$:

1. **Bayes-OAMP**, as described and implemented in Sections 4.1 and 4.2 above.
2. **Single-iterate Bayes-OAMP**, using instead the single-iterate posterior mean nonlinearities $u_{t+1}(f_0, \dots, f_t) = \mathbb{E}[U_* | F_t = f_t]$, with debiasing coefficients and state evolution parameters estimated as in Section 4.2.
3. **Gaussian Bayes-AMP**, using the debiasing coefficients and state evolution described in [39] for i.i.d. noise. We also estimate the signal strengths θ_k and initial spectral alignments $\mu_{\text{pca},k}^2$ from (3.5) using functions $G(\cdot)$ that correspond to the semicircle law for symmetric i.i.d. noise.

SYMMETRIC MODEL. We consider the symmetric model (3.1) with $n = 4000$, in the settings

- (Semicircle) $\mathbf{W} \sim \text{GOE}(n)$ has i.i.d. $\mathcal{N}(0, 1/n)$ entries above the diagonal, and the limit spectral law Λ is the semicircle distribution supported on $[-2, 2]$.
- (Uniform) $\mathbf{W} = \mathbf{O}^\top \mathbf{A} \mathbf{O}$ where $\mathbf{A}$ has i.i.d. $\text{Uniform}[-\sqrt{3}, \sqrt{3}]$ diagonal entries, and $\mathbf{O}$ is uniformly random.
- (Centered Beta) $\mathbf{W} = \mathbf{O}^\top \mathbf{A} \mathbf{O}$ where $\mathbf{A}$ has i.i.d. $\sqrt{80/3} \cdot (\text{Beta}(3, 1) - 3/4)$ diagonal entries, and $\mathbf{O}$ is uniformly random.

In all three settings, Λ is normalized to have mean $\kappa_1 = 0$ and variance $\kappa_2 = 1$. Figure 1 compares estimation error across AMP iterations, for an example with a rank-2 signal where $K' = K = 2$, the two signal strengths are $(\theta_1, \theta_2) = (2, 1.6)$, and the elements of the two signal vectors are drawn i.i.d. from the discrete three-point prior U_* defined by

$$\frac{1}{2} \delta_{(0,1)} + \frac{1}{4} \delta_{(\sqrt{2},-1)} + \frac{1}{4} \delta_{(-\sqrt{2},-1)}. \quad (4.2)$$

The error is defined by the subspace distance $\|\Pi_{\mathbf{U}_*} - \Pi_{\mathbf{U}_t}\|$ where $\Pi_{\mathbf{U}} \in \mathbb{R}^{n \times n}$ is the orthogonal projector onto the two-dimensional column span of $\mathbf{U}$. From the left panel, for GOE noise, we observe

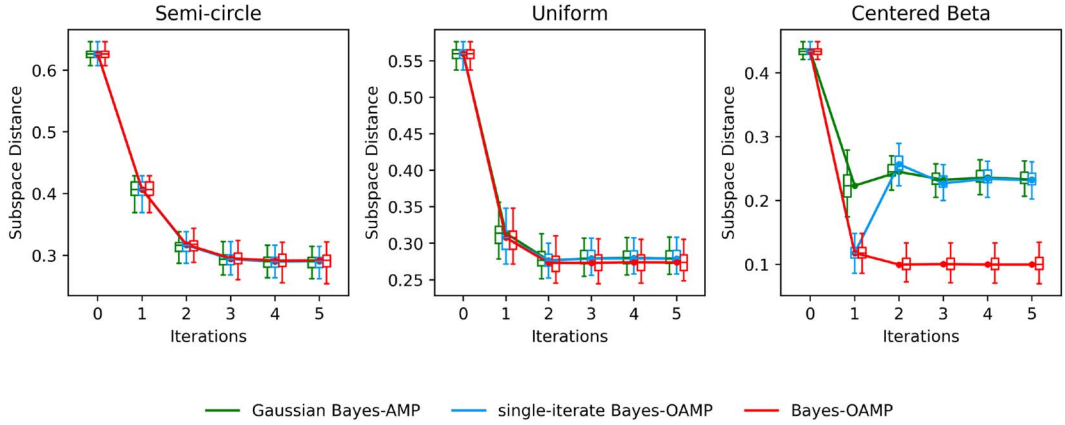


FIG. 1. Estimation errors for AMP iterates $\mathbf{U}_t$ in the symmetric spiked model with $n = 4000$, rank-2 signal, and signal prior $U_* \sim \frac{1}{2}\delta_{(0,1)} + \frac{1}{4}\delta_{(\sqrt{2},-1)} + \frac{1}{4}\delta_{(-\sqrt{2},-1)}$. Boxes indicate the $\{25, 50, 75\}$ -percentiles across 50 random trials, and whiskers indicate $1.5 \times$ inter-quartile range. Iteration 0 corresponds to the spectral initialization $\mathbf{U}_0$. The noise spectral distributions are (left) the semicircle law, (middle) Uniform $[-\sqrt{3}, \sqrt{3}]$, and (right) standardized Beta(3, 1).

that the three algorithms yield comparable per-iteration error, and there is negligible degradation in accuracy from estimating the spectral free cumulants and the more complex state parameters in Bayes-OAMP. In the middle panel, for Uniform noise spectrum, Gaussian Bayes-AMP remains reasonably robust to the misspecification of the noise distribution. In the right panel, for Centered Beta noise spectrum, we observe a significant improvement of Bayes-OAMP over the other two approaches, which both exhibit non-monotonic error across iterations.

Figure 2 depicts the accuracy of the state evolution predictions for the distribution of AMP iterates $\mathbf{F}_t$ in the Centered Beta example. Marginal histograms of the two columns of $\mathbf{F}_t$ are overlaid with the corresponding densities of F_t , computed using state evolution parameters (μ_t, Σ_t) empirically estimated as in Section 4.2. Note that these densities are exactly the ones assumed for posterior-mean denoising in (4.1), to obtain the next iterate $\mathbf{U}_{t+1}$. We observe a close agreement for Bayes-OAMP, and a large discrepancy with the state evolution predictions of Gaussian Bayes-AMP.

Figure 3 illustrates estimation error across different signal strengths, in an example having rank-1 signal and the same Centered Beta noise. Note that since the density of Beta(3, 1) does not decay to 0 at its right edge, the spectral phase transition point for this example is 0, and any positive signal strength is super-critical. We observe that Bayes-OAMP remains effective and improves over the spectral initialization for any positive signal strength. In contrast, single-iterate Bayes-OAMP exhibits the following behavior: below some critical signal strength, it diverges from the informative spectral initialization to an uninformative solution. This type of behavior was also reported in [38, Section 4] for a different prior. Our implementation of Gaussian Bayes-AMP uses the Cauchy-transform $G(\cdot)$ based on the semicircle law to infer θ , and thus is only applicable when the largest sample eigenvalue exceeds 2, corresponding roughly to $\theta \geq 1.6$ in this example. For these values of θ , we observe its accuracy to be close to that of single-iterate Bayes-OAMP.

RECTANGULAR MODEL. We consider the rectangular model

$$\mathbf{X} = \sum_{k=1}^{K'} \frac{\theta_k}{\sqrt{mn}} \mathbf{u}_*^k \mathbf{v}_*^{k\top} + \mathbf{W} \in \mathbb{R}^{m \times n}$$

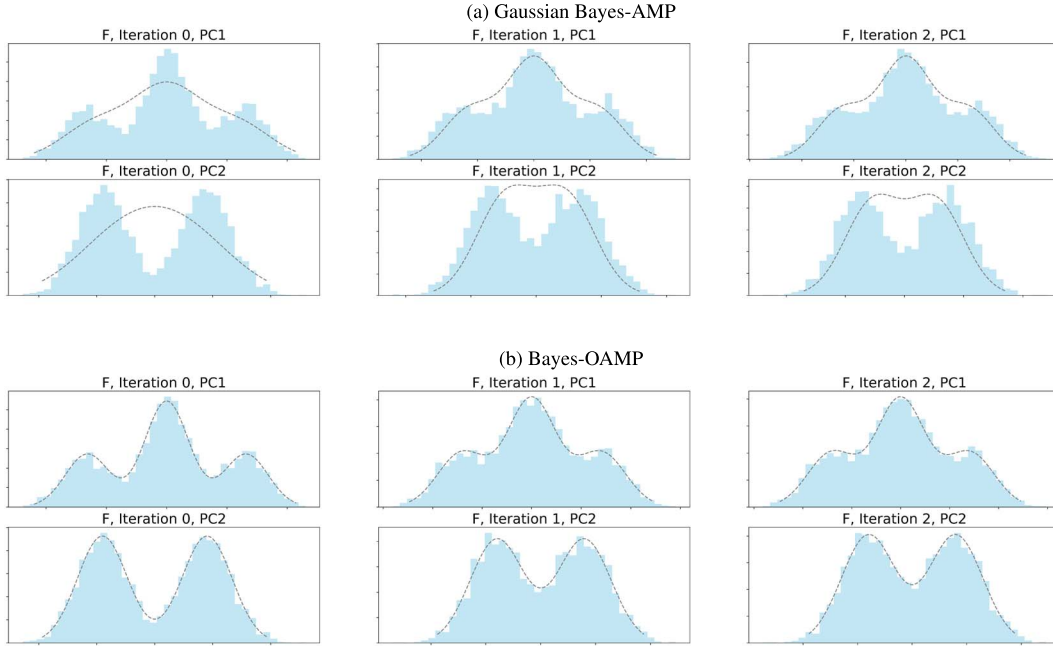


FIG. 2. Distributions of iterates $\mathbf{F}_0, \mathbf{F}_1, \mathbf{F}_2$ in the Centered Beta noise example of Fig. 1. Shown are histograms of the empirical distributions of the two columns of each iterate $\mathbf{F}_t$ (denoted PC1 and PC2), overlaid with the marginal density of the corresponding coordinate of the state evolution law $\mathcal{N}(\mu_t \cdot U_*, \Sigma_t)$. This density agrees closely for Bayes-OAMP, whereas a large discrepancy is observed for the state evolution prediction of Gaussian Bayes-AMP.

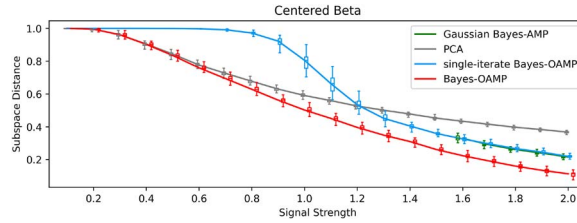


FIG. 3. Estimation errors for the sample eigenvector and three AMP algorithms, in a symmetric model with Centered Beta noise spectrum, $n = 4000$, rank-1 signal with prior $U_* \sim \frac{1}{2}\delta_{-1} + \frac{1}{2}\delta_1$, and signal strength θ varying from 0.1 to 2.0. The spectral ‘phase transition’ occurs at $\theta = 0$.

with $(m, n) = (3000, 4000)$, again with $K' = K = 2$, signal strengths $(\theta_1, \theta_2) = (2, 1.5)$, and the discrete three-point prior (4.2) for both U_* and V_* . We consider the settings

- (Marcenko–Pastur) $\mathbf{W}$ has i.i.d. $\mathcal{N}(0, 1/n)$ entries, and the limit singular value distribution Λ is the square-root of a Marcenko–Pastur law.
- (Uniform) $\mathbf{W} = \mathbf{O}^\top \mathbf{A} \mathbf{Q}$ where $\mathbf{A}$ has i.i.d. $\text{Uniform}[\sqrt{3/7}, 2\sqrt{3/7}]$ diagonal entries, and $(\mathbf{O}, \mathbf{Q})$ are uniformly random.

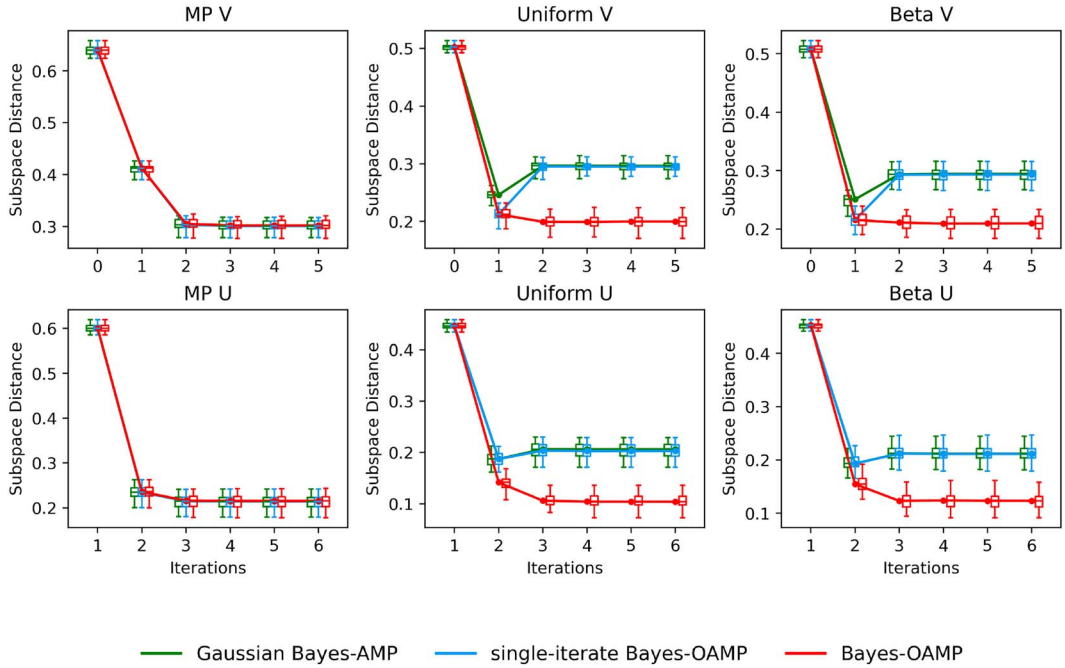


FIG. 4. Estimation errors for AMP iterates $\mathbf{U}_t$ and $\mathbf{V}_t$ in the rectangular spiked model with $(m, n) = (3000, 4000)$, rank-2 signal, signal priors $U_*, V_* \sim \frac{1}{2}\delta_{(0,1)} + \frac{1}{4}\delta_{(\sqrt{2},-1)} + \frac{1}{4}\delta_{(-\sqrt{2},-1)}$, and signal strengths $(\theta_1, \theta_2) = (2, 1.5)$. Iterates $\mathbf{V}_0$ and $\mathbf{U}_1$ correspond to the spectral initializations. The noise spectral distributions are (left) square-root of the Marcenko–Pastur law, (middle) Uniform $[\sqrt{3/7}, 2\sqrt{3/7}]$, and (right) the $\sqrt{5/3} \cdot \text{Beta}(3, 1)$ distribution.

- (Beta) $\mathbf{W} = \mathbf{O}^\top \mathbf{\Lambda} \mathbf{Q}$ where $\mathbf{\Lambda}$ has i.i.d. $\sqrt{5/3} \cdot \text{Beta}(3, 1)$ diagonal entries, and $(\mathbf{O}, \mathbf{Q})$ are uniformly random.

In all three settings, the limiting singular value distribution Λ of $\mathbf{W}$ is normalized so that $\mathbb{E}[\Lambda^2] = 1$.

The exact form of Bayes-OAMP algorithm for the rectangular model can be found in [Appendix C](#). [Figure 4](#) compares per-iteration estimation errors, where we again observe that Bayes-OAMP achieves the same error as Gaussian Bayes-AMP and single-iterate Bayes-OAMP for i.i.d. noise, but improves over the other procedures for the remaining noise settings.

5. Conclusion

We have developed AMP algorithms for both symmetric and rectangular spiked random matrix models in the context of orthogonally invariant noise. These algorithms extend those of [26] and [38] by allowing for matrix-valued AMP iterates, multivariate nonlinearities, and spectral initializations using super-critical sample eigenvectors or singular vectors of the observed data. We have derived the forms of the Onsager debiasing coefficients for a general class of such algorithms, and established rigorous Gaussian state evolutions for their iterates. These depend on the spectral distributions of the noise via their (symmetric or rectangular) free cumulants.

We developed one application of such algorithms to estimate the super-critical signal vectors in these models, by choosing Bayes posterior mean denoising functions as the nonlinearities in AMP. This Bayes-OAMP approach is similar to the algorithms of [39, 42], but applies the posterior mean conditional on all preceding AMP iterates, instead of only the single preceding iterate. Subsequent work of [5] suggests that alternating this posterior mean with identity nonlinearities may lead to further improvements of the Bayes-OAMP method, and potentially attain the Bayes-optimal estimation error. Computation of the posterior mean is enabled by the above state evolution, whose parameters may be estimated empirically. We observe in simulation that Bayes-OAMP can yield improved estimation accuracy over both the spectral initialization and single-iterate AMP algorithms, and can be more robust than standard AMP algorithms that are designed for white noise. These simulations suggest that the method may be sufficiently accurate and stable for practical use in PCA applications.

Acknowledgments

We would like to thank Marco Mondelli for a helpful discussion about [38].

Supplementary data

Supplementary data is available at *Information and Inference: a Journal of the IMA* online.

Funding

This research was supported in part by NSF DMS-1916198 and DMS-2142476.

REFERENCES

1. BAIK, J., BEN AROUS, G. & PÉCHÉ, S. (2005) Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Ann. Probab.*, **33**, 1643–1697.
2. BAIK, J. & SILVERSTEIN, J. W. (2006) Eigenvalues of large sample covariance matrices of spiked population models. *J. Multivariate Anal.*, **97**, 1382–1408.
3. BARBIER, J., KRZAKALA, F., MACRIS, N., MIOLANE, L. & ZDEBOROVÁ, L. (2019) Optimal errors and phase transitions in high-dimensional generalized linear models. *Proc. Natl. Acad. Sci. USA*, **116**, 5451–5460.
4. BARBIER, J., MACRIS, N., DIA, M. & KRZAKALA, F. (2020) Mutual information and optimality of approximate message-passing in random linear estimation. *IEEE Trans. Inform. Theory*, **66**, 4270–4303.
5. BARBIER, J., CAMILLI, F., MONDELLI, M. & SÁENZ, M. (2023) Fundamental limits in structured principal component analysis and how to reach them. *Proceedings of the National Academy of Sciences*, **120**, e2302028120.
6. BAYATI, M. & MONTANARI, A. (2011a) The LASSO risk for Gaussian matrices. *IEEE Trans. Inform. Theory*, **58**, 1997–2017.
7. BAYATI, M. & MONTANARI, A. (2011b) The dynamics of message passing on dense graphs, with applications to compressed sensing. *IEEE Trans. Inform. Theory*, **57**, 764–785.
8. BAYATI, M., LELARGE, M. & MONTANARI, A. (2015) Universality in polytope phase transitions and message passing algorithms. *Ann. Appl. Probab.*, **25**, 753–822.
9. BENAYCH-GEORGES, F. & NADAKUDITI, R. R. (2011) The eigenvalues and eigenvectors of finite, low rank perturbations of large random matrices. *Adv. Math.*, **227**, 494–521.
10. BERTHIER, R., MONTANARI, A. & NGUYEN, P.-M. (2020) State evolution for approximate message passing with non-separable functions. *Inf. Inference*, **9**, 33–79.

11. BOLTHAUSEN, E. (2014) An iterative construction of solutions of the TAP equations for the Sherrington–Kirkpatrick model. *Comm. Mth. Phys.*, **325**, 333–366.
12. BOLTHAUSEN, E. (2018) A Morita type proof of the replica-symmetric formula for SK. in *Proc. International Conference on Statistical Mechanics of Classical and Disordered Systems*, pp. 63–93. Switzerland: Springer.
13. BU, Z., KLUSOWSKI, J. M., RUSH, C. & SU, W. J. (2020) Algorithmic analysis and statistical estimation of SLOPE via approximate message passing. *IEEE Trans. Inform. Theory*, **67**, 506–537.
14. ÇAKMAK, B. & OPPER, M. (2020) A dynamical mean-field theory for learning in restricted Boltzmann machines. *J. Stat. Mech. Theory Exp.*, **10**, 103303.
15. CELENTANO, M., FAN, Z. & MEI, S. (2023) Local convexity of the TAP free energy and AMP convergence for -synchronization. *The Annals of Statistics*, **51**, 519–546.
16. CHEN, W.-K. & LAM, W.-K. (2021) Universality of approximate message passing algorithms. *Electron. J. Probab.*, **26**, 1–44.
17. DESHPANDE, Y. & MONTANARI, A. (2014) Information-theoretically optimal sparse PCA. in *Proc. IEEE International Symposium on Information Theory*, pp. 2197–2201. Honolulu, HI: IEEE.
18. DESHPANDE, Y., ABBE, E. & MONTANARI, A. (2017) Asymptotic mutual information for the balanced binary stochastic block model. *Inf. Inference*, **6**, 125–170.
19. DIA, M., MACRIS, N., KRZAKALA, F., LESIEUR, T. & ZDEBOROVÁ, L. (2016) Mutual information for symmetric rank-one matrix estimation: A proof of the replica formula. in *Proc. Advances in Neural Information Processing Systems*, pp. 424–432. Barcelona, Spain.
20. DING, J. & SUN, N. (2019) Capacity lower bound for the Ising perceptron. in *Proc. the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pp. 816–827. Phoenix, AZ.
21. DONOHO, D. L., MALEKI, A. & MONTANARI, A. (2009) Message-passing algorithms for compressed sensing. *Proc. Natl. Acad. Sci. USA*, **106**, 18914–18919.
22. DONOHO, D. L., MALEKI, A. & MONTANARI, A. (2010a) Message passing algorithms for compressed sensing: I. Motivation and construction. in *Proc. IEEE information theory workshop on information theory (ITW 2010, Cairo)*, pp. 1–5. Cairo, Egypt: IEEE.
23. DONOHO, D. L., MALEKI, A. & MONTANARI, A. (2010b) Message passing algorithms for compressed sensing: II. Analysis and validation. in *Proc. IEEE Information Theory Workshop on Information Theory (ITW 2010, Cairo)*, pp. 1–5. Cairo, Egypt: IEEE.
24. DONOHO, D. L., JAVANMARD, A. & MONTANARI, A. (2013) Information-theoretically optimal compressed sensing via spatial coupling and approximate message passing. *IEEE Trans. Inform. Theory*, **59**, 7434–7464.
25. DONOHO, D. & MONTANARI, A. (2016) High dimensional robust M-estimation: asymptotic variance via approximate message passing. *Probab. Theory Related Fields*, **166**, 935–969.
26. FAN, Z. (2022) Approximate message passing algorithms for rotationally invariant matrices. *The Annals of Statistics*, **50**, 197–224.
27. FAN, Z. & WU, Y. (in press) The replica-symmetric free energy for Ising spin glasses with orthogonally invariant couplings. *Probab. Theory Related Fields*.
28. FENG, O. Y., VENKATARAMANAN, R., RUSH, C., SAMWORTH, R. J., et al. (2022) A unifying tutorial on approximate message passing. *Foundations and Trends in Machine Learning*, **15**, 335–536.
29. JAVANMARD, A. & MONTANARI, A. (2013) State evolution for general approximate message passing algorithms, with applications to spatial coupling. *Inf. Inference*, **2**, 115–144.
30. KABASHIMA, Y. (2003) A CDMA multiuser detection algorithm on the basis of belief propagation. *Journal of Physics A: Mathematical and General*, **36**, 11111–11121.
31. KRZAKALA, F., MÉZARD, M., SAUSSET, F., SUN, Y. & ZDEBOROVÁ, L. (2012) Probabilistic reconstruction in compressed sensing: algorithms, phase diagrams, and threshold achieving matrices. *J. Stat. Mech. Theory Exp.*, **2012**, P08009.
32. LIU, L., HUANG, S. & KURKOSKI, B. M. (2021) Memory approximate message passing. in *Proc. IEEE International Symposium on Information Theory (ISIT)*, pp. 1379–1384. Melbourne, Australia: IEEE.
33. MA, J. & PING, L. (2017) Orthogonal AMP. *IEEE Access*, **5**, 2020–2033.

34. MALEKI, A. (2010) *Approximate message passing algorithms for compressed sensing* Ph.D. thesis. Stanford, CA: Stanford University.
35. MALEKI, A., ANITORI, L., YANG, Z. & BARANIUK, R. G. (2013) Asymptotic analysis of complex LASSO via complex approximate message passing (CAMP). *IEEE Trans. Inform. Theory*, **59**, 4290–4308.
36. MINGO, J. A. & SPEICHER, R. (2017) *Free probability and random matrices*, vol. **35**. New York, NY: Springer.
37. MONDELLI, M. & VENKATARAMANAN, R. (2021a) Approximate message passing with spectral initialization for generalized linear models. in *Proc. International Conference on Artificial Intelligence and Statistics*, pp. 397–405. San Diego, CA: PMLR.
38. MONDELLI, M. & VENKATARAMANAN, R. (2021b) PCA initialization for approximate message passing in rotationally invariant models. *Advances in Neural Information Processing Systems*, **34**, 29616–29629.
39. MONTANARI, A. & VENKATARAMANAN, R. (2021) Estimation of low-rank matrices via approximate message passing. *Ann. Statist.*, **49**, 321–345.
40. OPPER, M., ÇAKMAK, B. & WINTHER, O. (2016) A theory of solving TAP equations for Ising models with general invariant random matrices. *J. Phys. A.*, **49**, 114002.
41. PAUL, D. (2007) Asymptotics of sample eigenstructure for a large dimensional spiked covariance model. *Statist. Sinica*, **17**, 1617–1642.
42. RANGAN, S. & FLETCHER, A. K. (2012) Iterative estimation of constrained rank-one matrices in noise. in *Proc. IEEE International Symposium on Information Theory Proceedings*, pp. 1246–1250. Cambridge, MA: IEEE.
43. RANGAN, S., SCHNITER, P., FLETCHER, A. K. & SARKAR, S. (2019a) On the convergence of approximate message passing with arbitrary matrices. *IEEE Trans. Inform. Theory*, **65**, 5339–5351.
44. RANGAN, S., SCHNITER, P. & FLETCHER, A. K. (2019b) Vector approximate message passing. *IEEE Trans. Inform. Theory*, **65**, 6664–6684.
45. SCHNITER, P., RANGAN, S. & FLETCHER, A. K. (2016) Vector approximate message passing for the generalized linear model. in *Proc. 50th Asilomar Conference on Signals, Systems and Computers*, pp. 1525–1529. Pacific Grove, CA: IEEE.
46. SU, W., BOGDAN, M. & CANDÈS, E. (2017) False discoveries occur early on the lasso path. *Ann. Statist.*, **45**, 2133–2150.
47. SUR, P., CHEN, Y. & CANDÈS, E. J. (2019) The likelihood ratio test in high-dimensional logistic regression is asymptotically a rescaled chi-square. *Probab. Theory Related Fields*, **175**, 487–558.
48. TAKEUCHI, K. (2017) Rigorous dynamics of expectation-propagation-based signal recovery from unitarily invariant measurements. in *Proc. IEEE International Symposium on Information Theory (ISIT)*, pp. 501–505. Aachen, Germany: IEEE.
49. TAKEUCHI, K. (2019) A unified framework of state evolution for message-passing algorithms. in *Proc. IEEE International Symposium on Information Theory (ISIT)*, pp. 151–155. Paris, France: IEEE.
50. TAKEUCHI, K. (2020) Convolutional approximate message-passing. *IEEE Signal Processing Letters*, **27**, 416–420.
51. TAKEUCHI, K. (2021) Bayes-optimal convolutional AMP. *IEEE Transactions on Information Theory*, **67**, 4405–4428.
52. VENKATARAMANAN, R., KÖGLER, K. & MONDELLI, M. (2022) Estimation in rotationally invariant generalized linear models via approximate message passing. in *International Conference on Machine Learning*, pp. 22120–22144. Baltimore, MD: PMLR.
53. VILA, J., SCHNITER, P., RANGAN, S., KRZAKALA, F. & ZDEBOROVÁ, L. (2015) Adaptive damping and mean removal for the generalized approximate message passing algorithm. in *Proc. IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 2021–2025. South Brisbane, Australia: IEEE.
54. ZHONG, X., SU, C. & FAN, Z. (2022) Empirical Bayes PCA in high dimensions. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, **84**, 853–878.
55. ZIEMER, W. P. (2012) *Weakly differentiable functions: Sobolev spaces and functions of bounded variation*, vol. **120**. New York, NY: Springer Science & Business Media.